25900046

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCCSC100, MG2DSCECC101

- PYTHON PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Scheme Of Evaluation

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. What does it mean when we say Python is an interpreted language? [K] / [CO1]

   Python is an interpreted language, it reads python code line by line, translates each line into an intermediate form (often bytecode), and then executes that intermediate form directly. Explanation – 2 marks

2. Define Continue Statement. [K] / [CO2]

   The continue statement in Python is a control flow statement that skips the rest of the code inside the current iteration of a loop and proceeds to the next iteration. Explanation – 2 marks

3. Explain the purpose of the close() method in file handling. [U] / [CO3]

   close() method in file handling in Python is to release the system resources that were allocated when the file was opened. Explanation – 2 marks

4. Write an example of tuple unpacking. [U] / [CO3]

   Example – 2 marks

5. Define built-in functions. List 2 examples. [K] / [CO3]

   Definition – 1 mark. Any 2 examples – 0.5 marks each

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. How do you create a string in Python? [U] / [CO1]

Using Single Quotes

Using Double Quotes

Using Triple Quotes

Explanation – 4 marks, Example – 1 mark

7. How can we read multiple inputs from a user in Python? [K] / [CO1]

Explanation – 4 marks, Example – 1 mark

8. Explain the concept of while loop statement. [U] / [CO2]

Explanation – 4 marks, Example – 1 mark

9. Write Python script to test whether a given year is leap year or not [U] / [CO2]

```
y = 200
if y % 4 == 0:
   if y % 100 == 0:
      if y % 400 == 0:
         print("Leap year")
      else:
         print("Not a leap year")
   else:
      print("Leap year")
else:
   print("Not a leap year")
```

Program – 5 marks

10. Explain the importance of closing a file after performing file operations. [U] / [CO3]

Explanation – 5 marks

11. Write Python script to create a Dictionary and find the number of elements in it. [A] / [CO3]

```
my_dict = {
   "name": "Alice",
   "age": 30,
   "city": "New York",
   "occupation": "Engineer"
}
number_of_elements_my_dict = len(my_dict)
print("Dictionary 'my_dict':", my_dict)
print("Number of elements in 'my_dict':", number_of_elements_my_dict)
```

Program – 5 marks

12. Explain insert() and remove() methods of list with suitable examples of each. [U] / [CO3]

insert() – Explanation – 1.5 marks, Example – 1 mark
remove() – Explanation – 1.5 marks, Example – 1 mark

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Discuss the operators used in Python. Explain with example [U] / [CO1]

Explanation of 7 operators with example – 15 marks

Arithmetic Operators, Comparison Operators, Assignment Operators, Logical Operators, Identity Operators, Membership Operators, Bitwise Operators.

14. Explain the concept of lists in Python. Discuss different ways to create a list with examples. Explain at least five basic list operations and describe the list() function with relevant examples. [U] / [CO3]

Concept of lists in Python – 5 marks
Different ways to create a list with examples – 3 marks
Five basic list operations – 1 mark each
Description of list() function with relevant examples – 2 marks

[15x1 = 15]